

cryptology Assignment - 2

- ① Encrypt "CHASE YOUR DREAMS" using
- caesar cipher
 - shift cipher with key 7
 - ROT 13



A	B	C	D	E	F	G	H	I	J	K	L	M	N
O	P	Q	R	S	T	U	V	W	X	Y	Z		

a) caesar cipher $[K=3]$

$$C = E(P+3) \mod 26$$

CHASE

$$C : E(2+3) \mod 26$$

$$5 \mod 26 \rightarrow 5 = F$$

$$H : E(7+3) \mod 26$$

$$10 \mod 26 \rightarrow 10 = K$$

$$A : E(0+3) \mod 26$$

$$3 \mod 26 \rightarrow 3 = D$$

$$S : E(18+3) \mod 26$$

$$21 \mod 26 \rightarrow 21 = V$$

$$E : E(4+3) \mod 26$$

$$7 \mod 26 \Rightarrow 7 = H$$

YOUR

$$Y: E(24+3) \bmod 26$$

$$27 \bmod 26 = 1 \rightarrow B$$

$$O: E(14+3) \bmod 26$$

$$17 \bmod 26 = 17 \rightarrow R$$

$$U: E(20+3) \bmod 26$$

$$23 \bmod 26 = 23 \rightarrow X$$

$$R: E(17+3) \bmod 26$$

$$20 \bmod 26 = 20 \rightarrow U$$

DREAMS

$$D: E(3+3) \bmod 26$$

$$6 \bmod 26 \rightarrow 6 = G$$

$$R: E(17+3) \bmod 26$$

$$20 \bmod 26 \Rightarrow 20 = U$$

$$E: E(4+3) \bmod 26$$

$$7 \bmod 26 = 7 = H$$

$$A: E(0+3) \bmod 26$$

$$3 \bmod 26 = 3 = D$$

$$M: E(12+3) \bmod 26$$

$$15 \bmod 26 = 15 = P$$

$$S: E(18+3) \bmod 26$$

$$21 \bmod 26 = 21 = V$$

→ FKDVH BRXU GUHDPV

b) Shift cipher $K=7$

$$C = E(P+n) \bmod 26$$

CHASE

$$C: E(2+7) \bmod 26$$

$$9 \bmod 26 = 9 = J$$

$$H: C = E(7+7) \bmod 26$$

$$14 \bmod 26 = 14 : O$$

$$A: C = E(0+7) \bmod 26$$

$$7 \bmod 26 = 7 : H$$

$$S: C = E(18+7) \bmod 26$$

$$25 \bmod 26 = 25 : Z$$

$$E: C = E(4+7) \bmod 26$$

$$11 \bmod 26 = 11 : L$$

YOUR

$$Y: C = E(24+7) \bmod 26$$

$$31 \bmod 26 = 5 : F$$

$$O: C = E(14+7) \bmod 26$$

$$21 \bmod 26 = 21 : V$$

$$U: C = E(20+7) \bmod 26$$

$$27 \bmod 26 = 1 : B$$

$$R: C = E(17+7) \bmod 26$$

$$24 \bmod 26 = 24 : Y$$

DREAMS :

$$D: C = E(3+7) \bmod 26$$

$$10 \bmod 26 = 10 : K$$

$$R: C = E(17+7) \bmod 26$$

$$24 \bmod 26 = 24 : Y$$

$$E: C = E(4+7) \bmod 26$$

$$11 \bmod 26 = 11 : L$$

$$A: C = E(0+7) \bmod 26$$

$$7 \bmod 26 = 7 : H$$

$$M: C = E(12+7) \bmod 26$$

$$19 \bmod 26 = 19 : T$$

$$S: C = E(18+7) \bmod 26$$

$$25 \bmod 26 = 25 : Z$$

JOH2L

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classmate
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c) ROT 13 $K = 13$

$$C = E(P + 13) \bmod 26$$

CHASE

$$C : C = E(2 + 13) \bmod 26$$

$$15 \bmod 26 = 15 \rightarrow P$$

$$H : C = E(7 + 13) \bmod 26$$

$$20 \bmod 26 = 20 \rightarrow U$$

$$A : C = E(0 + 13) \bmod 26$$

$$13 \bmod 26 = 13 \rightarrow N$$

$$S : C = E(18 + 13) \bmod 26$$

$$31 \bmod 26 = 5 \rightarrow F$$

$$E : C = E(4 + 13) \bmod 26$$

$$17 \bmod 26 = 17 \rightarrow R$$

YOUR

$$Y : C = E(24 + 13) \bmod 26$$

$$37 \bmod 26 = 11 \rightarrow L$$

$$O : C = E(14 + 13) \bmod 26$$

$$27 \bmod 26 = 1 \rightarrow B$$

$$U : C = E(20 + 13) \bmod 26$$

$$33 \bmod 26 = 7 \rightarrow H$$

$$R : C = E(17 + 13) \bmod 26$$

$$30 \bmod 26 = 4 \rightarrow E$$

DREAMA

$$D : C = E(3 + 13) \bmod 26$$

$$16 \bmod 26 = 16 \rightarrow Q$$

$$R : C = E(17 + 13) \bmod 26$$

$$30 \bmod 26 = 4 \rightarrow E$$

$$E : C = E(4 + 13) \bmod 26$$

$$17 \bmod 26 = 17 \rightarrow R$$

$$A: C = E(0 + 13) \bmod 26$$

$$13 \bmod 26 = 13 = N$$

$$M: C = E(12 + 13) \bmod 26$$

$$25 \bmod 26 = 25 = Z$$

$$S: C = E(18 + 13) \bmod 26$$

$$31 \bmod 26 = 5 = F$$

PUNFR

LBHE

QERNZF

② Decrypt

a) "WRWDOOB VHFUHW PHVVDJH" using caesar cipher
 $k = 3$

$$P = D(C - 3) \bmod 26$$

WRWDOOB

$$W = 22 \rightarrow (22 - 3) \bmod 26$$

$$19 \bmod 26 = 19 = T$$

$$R = (17 - 3) \bmod 26$$

$$14 \bmod 26 = 14 = O$$

$$W = 19 = T$$

$$D = (3 - 3) \bmod 26$$

$$0 \bmod 26 = A$$

$$O = (14 - 3) \bmod 26$$

$$11 \bmod 26 = L$$

$$O = L$$

$$B = (1 - 3) \bmod 26$$

$$-2 \bmod 26 = 24 = Y$$

VH FVHW

$$V = (21 - 3) \bmod 26$$

$$18 \bmod 26 = S$$

$$H = (7 - 3) \bmod 26$$

$$4 \bmod 26 = E$$

$$F = (5 - 3) \bmod 26$$

$$2 \bmod 26 = C$$

$$V = (20 - 3) \bmod 26$$

$$17 \bmod 26 = R$$

$$H = (7 - 3) \bmod 26$$

$$4 \bmod 26 = E$$

$$W = (22 - 3) \bmod 26$$

$$19 \bmod 26 = T$$

PHVVVDJH

$$P = (15 - 3) \bmod 26 = M$$

$$H = (7 - 3) \bmod 26 = E$$

$$V = (21 - 3) \bmod 26 = S$$

$$V = (21 - 3) \bmod 26 = S$$

$$D = (3 - 3) \bmod 26 = A$$

$$J = (9 - 3) \bmod 26 = G$$

$$H = (7 - 3) \bmod 26 = E$$

TOTALLY SECRET MESSAGE

b) 'SJL IBTW YNRJ' using shift cipher, 5

$$P = (C - 5) \bmod 26$$

SJL

$$S = (18 - 5) \bmod 26 = M$$

$$J = (9 - 5) \bmod 26 = E$$

$$I = (8-5) \bmod 26 = D$$

IBTW

$$I = (8-5) \bmod 26 = D$$

$$B = (1-5) \bmod 26 = -4 \bmod 26 = 22 = W$$

$$T = (19-5) \bmod 26 = 14 \bmod 26 = O$$

$$W = (22-5) \bmod 26 = R$$

YNRJ

$$Y = (24-5) \bmod 26 = T$$

$$N = (13-5) \bmod 26 = I$$

$$R = (17-5) \bmod 26 = M$$

$$J = (9-5) \bmod 26 = E$$

NED DWOR TIME

c) "FRAC ZBER PBSSRR" using ROT 13.

$$P = D(C-13) \bmod 26$$

- FRAC;

$$F = (5-13) \bmod 26 = -8 \bmod 26 = 18 = S$$

$$R = (17-13) \bmod 26 = 4 \bmod 26 = E$$

$$A = (0-13) \bmod 26 = -13 \bmod 26 = N$$

$$Q = (16-13) \bmod 26 = 3$$

ZBER

$$Z = (25-13) \bmod 26 = M$$

$$B = (1-13) \bmod 26 = -12 \bmod 26 = 14 = O$$

$$E = (4-13) \bmod 26 = -9 \bmod 26 = 17 = R$$

$$R = (17-13) \bmod 26 = 4 = E$$

PBSSRR

$$P = (15 - 13) \bmod 26 = C$$

$$B = (1 - 13) \bmod 26 = D$$

$$S = (18 - 13) \bmod 26 = F$$

$$S = (18 - 13) \bmod 26 = F$$

$$R = (17 - 13) \bmod 26 = E$$

$$R = E$$

SEND MORE COFFEE